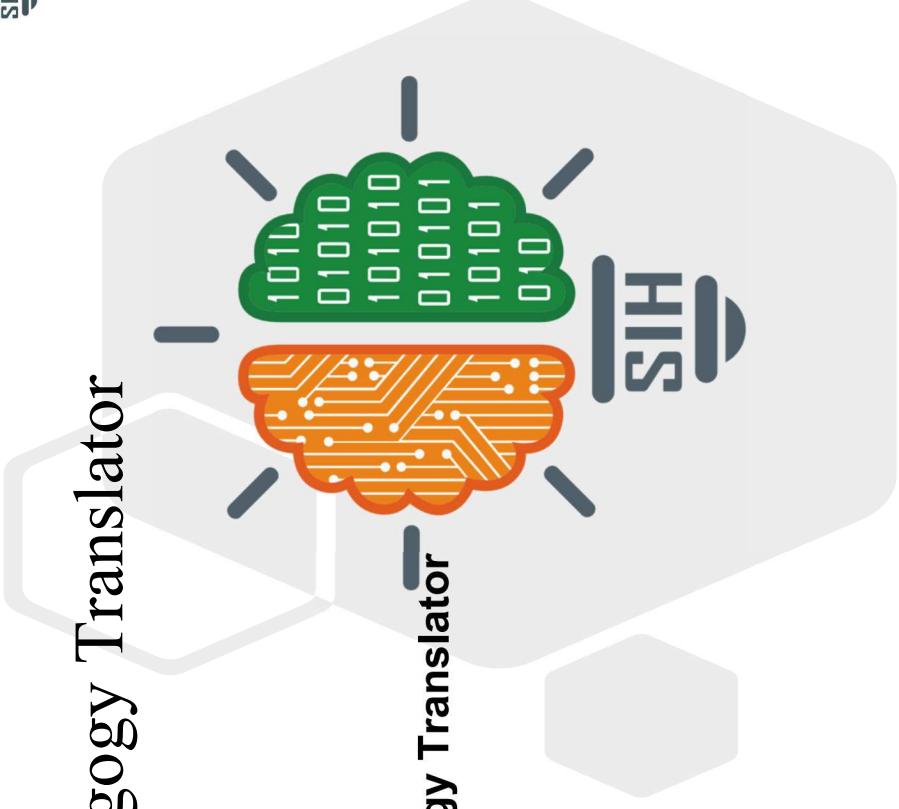


SMART INDIA HACKATHON 2026



Vernacular Pedagogy Translator



- Problem Statement ID – SIH26042
- Problem Statement Title- Vernacular Pedagogy Translator
- Theme- Education / Language & Learning
- PS Category- Software
- Team ID- 52
- Team Name- 404_FOUNDERS

Vernacular Pedagogy Translator

• Proposed Solution - Vernacular Pedagogy Translator

We are making an AI-Powered Vernacular Pedagogy Translate that converts English educational content into a regional language in a simple and easy to understand form. The system uses an LLM for translation and simplification and a Text to speech (TTS) API to read the translate lesson aloud.

Many students, especially in rural and non-English-speaking communities, face difficulties understanding educational content available mainly in English. Our solution removes the language barrier by providing learning material in the learner's familiar regional language and audio format. This makes education more accessible, understandable, and inclusive, even for learners with limited English proficiency or reading difficulties.

The key innovation is combining translation + educational simplification + regional-language voice output in a single platform.

TECHNICAL APPROACH



- **Technologies to be used**

Frontend: HTML, CSS, JavaScript / React

Backend: Python with Flask/FastAPI

AI/LLM: LLM API for translation and content simplification

Translation: Multilingual/Indian-language translation API

Text-to-Speech: TTS API for regional-language audio

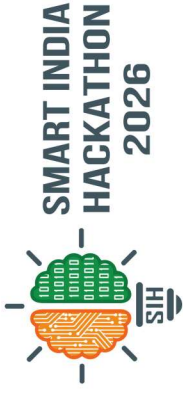
- **Methodology**

User Input → English Lesson → Text Processing → AI/LLM → Translate & Simplify → Regional Language Text → TTS → Audio Output → Learner

Working Prototype

The user enters or uploads an English lesson and selects a regional language. The backend sends the content to the LLM, which translates and simplifies it. The resulting regional-language text is displayed and passed to the TTS API to generate speech. The learner can then read and listen to the simplified lesson.

FEASIBILITY AND VIABILITY



1. Feasibility of the Idea

Uses existing LLM, translation, and Text-to-Speech APIs.

Can be developed as a lightweight web application.

Requires minimal hardware and can run on smartphones, tablets, or computers.

The basic MVP can be implemented within the 3-hour hackathon

2. Potential Challenges and Risks

Accuracy of translation for regional languages.

AI may sometimes generate incorrect or overly complex translations.

TTS support and pronunciation may vary across languages.

Internet/API dependency may affect performance

3. Strategies to Overcome Challenges

Use reliable multilingual AI/translation models and APIs.

Apply prompts that specifically instruct the AI to use simple, learner-friendly language.

Validate the translated output before generating audio.

Provide multiple language options and fallback mechanisms.

Optimize API calls and keep the MVP focused on short educational content.

IMPACT AND BENEFITS

1. Potential Impact on Target Audience
Helps non-English-speaking students understand educational content in their native language. Improves learning accessibility for rural and regional learners. Supports students who struggle with English-based textbooks and lessons. Audio output helps learners with reading difficulties and promotes inclusive education. Can be useful for students, teachers, and educational institutions.
2. Benefits of the Solution Breaks language barriers in education. Provides simple and easy-to-understand explanations. Enables learning through both text and audio. Supports multiple Indian regional languages. Saves time by automatically translating and simplifying lessons. Makes digital education more accessible, inclusive, and learner-friendly.

LINKS ARE USED IN

- <https://vernacular-classroom-translator--jayesh210507.replit.app>
- perplexity
- Replit
- Gemini
- Google
- Note pad
- Open API redmi